



Metropolitan University, Sylhet

Department of Computer Science & Engineering

Lab Report – 01: Greedy Algorithm (Fractional Knapsack)

Course Title: Algorithm Design and Analysis Lab

Course Code: CSE 232

Submitted To:

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problem:

Greedy Algorithm (Fractional Knapsack)

CODE :

```
// Fractional Knapsack using Greedy Method
// Name: Mahdi Hasan Shuvo | ID: 251-115-030 | Batch: CSE 62nd (A)

#include<iostream>
#include<vector>
#include<algorithm>
using namespace std;
struct Items
{
    int weight;
    int profit;
    float pw_ratio;
};

bool compare(Items a, Items b)
{
    return a.pw_ratio > b.pw_ratio;
}

int main()
{
    int n;
    cout << "Enter number of total item: ";
    cin >> n;

    vector <Items> item(n);

    for (int i = 0; i < n; i++)
    {
        cout << "weight-" << i+1 << ": ";
        cin >> item[i].weight;
        cout << "profit-" << i+1 << ": ";
        cin >> item[i].profit;

        item[i].pw_ratio = (float)item[i].profit/item[i].weight;
    }

    sort(item.begin(), item.end(), compare);

    float max_profit = 0;
```

```

int capacity;
cout << "Enter capacity: ";
cin >> capacity;

int remain_capacity = capacity;

for (int i = 0; i < n; i++)
{
    if (remain_capacity == 0)
        break;

    if (item[i].weight <= remain_capacity)
    {
        max_profit += item[i].profit;
        remain_capacity -= item[i].weight;
    }

    else
    {
        max_profit += remain_capacity * item[i].pw_ratio;
        remain_capacity = 0;
    }

}

cout << "\nMaximum possible profit: " << max_profit << endl;

return 0;
}

```

Output Output Output

OUTPUT:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Enter number of total item: 4
● weight-1: 11
  profit-1: 22
  weight-2: 12
  profit-2: 23
  weight-3: 14
  profit-3: 25
  weight-4: 15
  profit-4: 27
Enter capacity: 11

Maximum possible profit: 22
○ PS G:\CSE-62\AD&A> |
```